Prompt Design in the Conquer Phase

A structured prompt template for category selection with controlled output behavior

1 Input

For each sub-problem, run one MLLM inference

(a) Query image $\mathbf{x}$



(b) Candidate category list $\mathcal{G}_i^{(t)}$

headphone,
wild_cat,
panda,
starfish,
rhino

2 Prompt Template

Designed to enforce structural and semantic consistency

1 Task instruction
Please directly identify the category name of the image from the candidate list.

2 Decision rule
Always choose the most likely category whenever possible; output 'None' only if the image clearly does not belong to any category in the list.

3 Output constraint
Do not output sentences or explanations; output only the category name exactly as listed (preserve case and singular/plural form).

4 One-shot exemplar
Q: There are 3 categories, [Leopards, wild_cat, electric_guitar]. Which category does the image belong to?
A: wild_cat

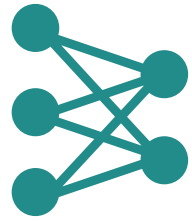
**Q: There are { $\left|\mathcal{G}_i^{(t)}\right|$ } categories, [','.join($\mathcal{G}_i^{(t)}$)].
Which category does the image belong to? A:**

Query image

Prompt template

3 MLLM Inference

Multimodal Large Language Model



Conquer phase inference

4 Constrained Output

Output: single category name

panda

choose most probable class

OR

None

only when no listed class fits


Image

+


Candidate list $\mathcal{G}_i^{(t)}$

+


Prompt template


MLLMs


Category name

Consistency
Shared template yields stable structure across sub-problems.

Anti-laziness
Discourages overuse of 'None' by preferring the most likely class.

Format fidelity
Keeps exact casing and singular/plural form from the candidate list.

One-shot guidance
Example steers the model toward the desired response format.